2025.11.3 (Derivatives of Distribution)

Motivation: $f\in C^1(\Omega)\Rightarrow f\in L^1_{loc}(\Omega)$, $\frac{\partial f}{\partial x_i}\in L^1_{loc}(\Omega)$ $i=1\cdots n$.

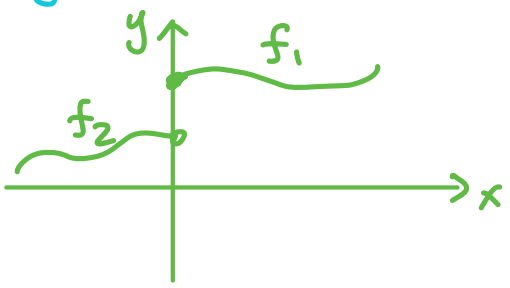
$\forall \phi\in \mathcal{D}(\Omega)=C_c^\infty(\Omega)$, $\langle \frac{\partial f}{\partial x_i},\phi\rangle=\int_\Omega \frac{\partial f}{\partial x_i}\cdot\phi\,dx=-\int_\Omega f\cdot\frac{\partial \phi}{\partial x_i}dx=-\langle f,\frac{\partial\phi}{\partial x_i}\rangle$

Def1 $\forall f\in\mathcal{D}'(\Omega)$, define distributional derivatives $D_i f$ of f by $\langle D_i f,\phi\rangle=-\langle f,\frac{\partial\phi}{\partial x_i}\rangle$ $\quad f_{x_i}=\frac{\partial f}{\partial x_i}=\partial_{x_i} f$

Fact $D_i f\in\mathcal{D}'(\Omega)$ In general, $\forall \alpha\in\mathbb{Z}^n$, $\alpha_i\geq 0$, define distributional derivative $D^\alpha f$ by: $D^\alpha f=\frac{\partial^{\alpha_1+\cdots+\alpha_n} f}{\partial^{\alpha_1}x_1\cdots\partial^{\alpha_n}x_n}$, $|\alpha|=\alpha_1+\cdots+\alpha_n$.

$\langle D^\alpha f,\phi\rangle=(-1)^{|\alpha|}\langle f,D^\alpha\phi\rangle$ $\forall\phi\in\mathcal{D}(\Omega)$. $\Rightarrow D^\alpha f\in\mathcal{D}'(\Omega)$

e.g.1 $f_1\in C^1[0,\infty)$, $f_2\in C^1(-\infty,0]$ Consider: $f(x)=\begin{cases} f_1(x), & x\in[0,\infty)\\ f_2(x), & x\in(-\infty,0)\end{cases}$



$f\in L^1_{loc}(\mathbb{R})$

Q: Df in sense of distribution.

$\langle Df,\phi\rangle=-\langle f,D\phi\rangle=-\int_{-\infty}^{\infty} f\cdot\phi'(x)dx=-\int_{-\infty}^{0} f_2(x)\cdot\phi'(x)dx-\int_0^{\infty} f_1(x)\phi'(x)dx$

$=\int_{-\infty}^{\infty} f_2'(x)\phi(x)dx-f_2(0^-)\phi(0^-)+f_2(-\infty)\phi(-\infty)+\int_0^{\infty} f_1'(x)\phi(x)dx-f_1(0^+)\phi(\infty)$

$=\int_{-\infty}^{\infty} f'\phi\,dx+[f_1(0)-f_2(0)]\phi(0)=\langle f',\phi\rangle+[f_1(0)-f_2(0)]\langle\delta_0,\phi\rangle+f_1(0^+)\phi(0^+)$

$\Rightarrow f'(x)=\begin{cases} f_1', & x\in(0,\infty)\\ f_2', & x\in(-\infty,0)\\ \text{any value}, & x=0\end{cases}$ $\Rightarrow Df=f'+(f_1(0)-f_2(0))\delta_0$

Df: distr's; f': classical; $(f_1(0)-f_2(0))\delta_0$: jump

Distributional solutions of PDE

Operator: $u\in\mathcal{D}'(\Omega)\mapsto Lu=\sum_{|\alpha|=0}^{m} A_\alpha(x)\underbrace{D^\alpha u}_{\in\mathcal{D}'(\Omega)}$, $A_\alpha(x)\in C^\infty(\Omega)$ $\quad\in\mathcal{D}'(\Omega)$

$\forall\phi\in\mathcal{D}(\Omega)$

$\langle Lu,\phi\rangle=\langle\sum_{|\alpha|=0}^{m} A_\alpha(x)D^\alpha u,\phi\rangle=\sum_{|\alpha|=0}^{m}\langle A_\alpha(x)D^\alpha u,\phi\rangle=\sum_{|\alpha|=0}^{m}\langle D^\alpha u,A_\alpha(x)\phi\rangle$

$= \sum_{|\alpha|=0}^{m} (-1)^{|\alpha|} \langle u, D^\alpha(A_\alpha(x)\phi) \rangle = \langle u, \sum_{|\alpha|=0}^{m} (-1)^{|\alpha|} D^\alpha(A_\alpha(x)\phi) \rangle := \langle u, L^*\phi \rangle$

$\Rightarrow \langle Lu, \phi \rangle = \langle u, L^*\phi \rangle \quad L: \mathcal{D}'(\Omega) \to \mathcal{D}'(\Omega)$

$L^*: \mathcal{D}(\Omega) \to \mathcal{D}(\Omega)$ ↓ dual operator.

e.g.2 $L = \Delta$, then $L^* = \Delta$ $\langle \Delta u, \phi \rangle = \langle u, \Delta\phi \rangle$

↘ Symmetric. ↙

Def3 PDE: $Lu = f$, $f \in \mathcal{D}'(\Omega)$. If $v \in \mathcal{D}'(\Omega)$ satisfies, $Lv = f$ in the sense of distribution, i.e. $\langle Lv, \phi \rangle = \langle f, \phi \rangle \ \forall \phi \in \mathcal{D}(\Omega)$. Then we say v is a distributional solution of PDE.

Remark: classical solution must be distributional solution.

Def4 A distributional solution of $-\Delta u = \delta_y$, in $\mathbb{R}^n$, $y \in \mathbb{R}^n$ fixed. is called a fundamental solution (f.c). → No uniqueness.

u is f.c $\Rightarrow u+c$ is f.c.

$-\langle \Delta u, \phi \rangle = -\langle \delta_y, \phi \rangle$; $-\langle \Delta(u+c), \phi \rangle = -\langle u+c, \Delta\phi \rangle$

$\| \quad -\langle u, \Delta\phi \rangle = -\langle u, \Delta\phi \rangle - \langle c, \Delta\phi \rangle = -\langle u, \Delta\phi \rangle$

e.g.3 F.S $u(x) = \begin{cases} -\frac{1}{2\pi} \ln|x-y|, & n=2 \\ \frac{1}{(n-2)w_n} \frac{1}{|x-y|^{n-2}}, & n \geq 3 \end{cases}$ w_n = area of unit sphere in $\mathbb{R}^n$

$\Delta u = 0$.

$u(x)$ is a F.S. of $-\Delta u = \delta_y$.

Pf: Step1: Check $u \in L^1_{loc}(\Omega)$

$u(x) = u(r)$, $r = |x-y|$, $u = \begin{cases} c \ln r, & n=2 \\ \frac{c_n}{r^{n-2}}, & n \geq 3 \end{cases}$

$n=2$: $\lim_{r \to 0^+} \frac{\ln r}{r^{-\frac{1}{2}}} = \lim_{r \to 0^+} \frac{1}{r} \cdot \frac{1}{-\frac{1}{2} r^{-3/2}} = \lim_{r \to 0^+} \frac{2r}{-1} = 0.$

$\int_{B_1(y)} r^{-\frac{1}{2}} dr = \frac{4\pi}{3} < \infty$

$n\geq 3$: $\int_{B_1(y)} \frac{1}{|x|^{n-2}}\,dx = \int_{\partial B_1}\int_0^1 r\,dr\,dS < \infty$. $u \notin L^1(\Omega)$

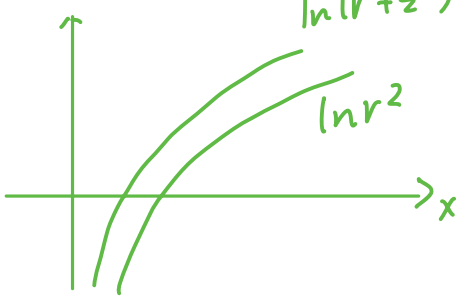
Step2: $\langle -\Delta u, \phi\rangle = \langle \delta_y, \phi\rangle$, $\phi \in C_c^\infty(\mathbb{R}^n)$. i.e. $-\int_{\mathbb{R}^n} u\Delta\phi\,dx = \phi(y)$.

Regularize u as: $u_\varepsilon(x) = \begin{cases} \frac{c}{2}\ln(r^2+\varepsilon^2), & n=2 \\ \frac{c_n}{(r^2+\varepsilon^2)^{\frac{n-2}{2}}}, & n\geq 3 \end{cases}$ $\Rightarrow \forall \varepsilon>0,\ u_\varepsilon \in C^\infty(\mathbb{R}^n)$.

$\int_{\mathbb{R}^n} u_\varepsilon \Delta\phi\,dx = \int_{\mathbb{R}^n} \Delta u_\varepsilon\,\phi\,dx$

W.T.S $\begin{cases} (a)\int_{\mathbb{R}^n} u_\varepsilon\Delta\phi\,dx \to \int_{\mathbb{R}^n} u\Delta\phi\,dx \quad \varepsilon\to 0 & \text{By DCT} \\ (b)\int_{\mathbb{R}^n} \Delta u_\varepsilon\,\phi\,dx \to \int_{\mathbb{R}^n} \Delta u\,\phi\,dx \quad \varepsilon\to 0 & \text{By Spicky Thm} \end{cases}$

$n=2$: (a) $\forall \phi \in D(\mathbb{R}^n)$. Choose $R>0$ s.t. $\operatorname{supp}\phi \subset B_R(y)$



On $B_{\frac12}(y)$, $r^2 < r^2+\varepsilon^2 < 1 \Rightarrow 0 < \frac{c}{2}\ln(r^2+\varepsilon^2) < \frac{c}{2}\ln r^2$

$\Rightarrow 0 < u_\varepsilon(r) < u(r)$

$\Rightarrow |u_\varepsilon(x)\Delta\phi| \leq |u(x)\Delta\phi| \in L^1(B_{\frac12}(y))$

By DCT, $\lim_{\varepsilon\to 0}\int_{B_{\frac12}(y)} u_\varepsilon\Delta\phi\,dx = \int_{B_{\frac12}(y)} u\Delta\phi\,dx$.

On $B_R(y)\setminus B_{\frac12}(y)$, $|u_\varepsilon(x)| \leq \frac{|c|}{2}\max\{|\ln\frac{1}{4}|, \ln(R+1)\} := M$

$|u_\varepsilon(x)\Delta\phi| \leq M\|\Delta\phi\|_{L^\infty(\mathbb{R}^2)} \leq \widetilde{M} \in L^1(B_R(y)\setminus B_{\frac12}(y))$

By BCT / DCT, $\lim_{\varepsilon\to 0}\int_{B_R(y)\setminus B_{\frac12}(y)} u_\varepsilon\Delta\phi\,dx = \int_{B_R(y)\setminus B_{\frac12}(y)} u\Delta\phi\,dx$

(b) W.T.S $\begin{cases} (i)\ \|-u_\varepsilon\|_{L^1(\mathbb{R}^2)} \leq M_3 \quad \forall \varepsilon>0. \\ (ii)\ \int_{\mathbb{R}^n} -\Delta u_\varepsilon\,dx \to 1 \quad \text{as } \varepsilon\to 0 \\ (iii)\ \forall \delta>0 \int_{|x-y|\geq\delta} |-\Delta u_\varepsilon|\,dx \to 0 \quad \text{as } \varepsilon\to 0 \end{cases}$

$\Delta u_\varepsilon = (u_\varepsilon)_{rr} + \frac{1}{r}(u_\varepsilon)_r = \frac{2c\varepsilon^2}{(r^2+\varepsilon^2)^2}$, $-\Delta u_\varepsilon > 0$ in $\mathbb{R}^2$

$\int_{\mathbb{R}^2} -\Delta u_\varepsilon\,dx = -2c\varepsilon^2\int_0^{2\pi}\int_0^\infty \frac{r}{(r^2+\varepsilon^2)^2}\,dr\,d\theta = c\varepsilon^2\cdot 2\pi\,\frac{1}{r^2+\varepsilon^2}\Big|_0^\infty = -2\pi c = 1$

$\int_{|x-y|\geq\delta} |-\Delta u_\varepsilon|\,dx = \frac{-c\varepsilon^2 2\pi}{\delta^2+\varepsilon^2} \to 0\ (\varepsilon\to 0)$ By Spiky Thm.

$n\geq 3$: (a) $|u_\varepsilon(x)\Delta\phi| \leq |u(x)\Delta\phi(x)| \in L^1(B_R(y))$

(b) $\Delta u_\varepsilon = -c_n n(n-2)\varepsilon^2 (r^2+\varepsilon^2)^{-\frac{n+2}{2}}$

$$\int_{\mathbb{R}^n} -\Delta u_\varepsilon \, dx = n(n-2) c_n \int_{\mathbb{R}^n} \frac{\varepsilon^2}{(r^2+\varepsilon^2)^{\frac{n}{2}+1}} dx = (n-2) c_n w_n = 1$$

$$\int_{|x-y|>\delta} |-\Delta u_\varepsilon| \, dx = n(n-2) c_n \cdot w_n \varepsilon^2 \int_\delta^\infty \frac{r^{n-1}}{(r^2+\varepsilon^2)^{\frac{n}{2}+1}} dr < n\varepsilon^2 \int_\delta^\infty r^{-3} dr$$

$\to 0 \ (\varepsilon \to 0)$.

Let $\Gamma(x-y) = \begin{cases} -\frac{1}{2\pi} \ln|x-y|, & n=2 \\ \frac{1}{(n-2)w_n |x-y|^{n-2}}, & n \geq 3 \end{cases}$ $\Rightarrow \int_{\mathbb{R}^n} \Gamma(x-y)(-\Delta\phi) dx = \phi(y)$

$\Gamma * (-\Delta\phi) = \phi(y)$

Thm For Poisson eq. $-\Delta u = f(x)$. $\forall f \in C_0^2(\mathbb{R}^n)$, $u(x) = \int_{\mathbb{R}^n} \Gamma(x-y) f(y) dy$ $\in C^2(\mathbb{R}^n)$ satisfies Poisson eq.

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Thm For Poisson eq. $-\Delta u = f(x)$. $\forall f \in C_0^2(\mathbb{R}^n)$, $u(x) = \int_{\mathbb{R}^n} \Gamma(x-y) f(y) dy$ $\in C^2(\mathbb{R}^n)$ satisfies Poisson eq. in the sense of classical.

Pf: • $u(x)$ is well-defined.

$\Gamma(x-\cdot) \in L^1_{loc}(\mathbb{R}^n)$ & $f \in C_c^2(\mathbb{R}^n) \Rightarrow u(x)$ integrable.

• $\forall$ fixed $x \in \mathbb{R}^n$, $u(x) \overset{z=x-y}{=\!=} \int_{\mathbb{R}^n} \Gamma(z) f(x-z) dz = \int_{\mathbb{R}^n} \Gamma(y) f(x-y) dy$

$$\frac{u(x+he_i)-u(x)}{h} = \int_{\mathbb{R}^n} \Gamma(y) \frac{f(x+he_i-y)-f(x-y)}{h} dy = \int_{\mathbb{R}^n} \Gamma(y) \frac{\partial f}{\partial x_i}(x+se_i-y) dy$$

$(0<s<h)$. By DCT, $f \in C_0^2(\mathbb{R}^n)$, $\exists R = R(x)$ s.t. $\operatorname{supp} f \subset B_R(x)$

$|\Gamma(y) \frac{\partial f}{\partial x_i}(x+se_i-y)| \leq |\Gamma(y)| \, \|\frac{\partial f}{\partial x_i}\|_{L^\infty(\mathbb{R}^n)} \mathbb{1}_{B_{R+1}(x)}$ $\forall s$ small.

$\to \in L^1(\mathbb{R}^n)$

Let $h \to 0^+$. $u_{x_i} = \int_{\mathbb{R}^n} \Gamma(y) \frac{\partial f}{\partial x_i}(x-y) dy$

• Continuity of u_{x_i}. Take any $x^k \to x$ $(k \to \infty)$, $\frac{\partial u}{\partial x_i}(x^k) \to \frac{\partial u}{\partial x_i}(x)$.

$u_{x_i}(x^k) - u_{x_i}(x) = \int_{\mathbb{R}^n} \Gamma(y) \left(\frac{\partial f}{\partial x_i}(x^k-y) - \frac{\partial f}{\partial x_i}(x-y)\right) dy$. By DCT

$|\Gamma(y)\left(\frac{\partial f}{\partial x_i}(x^k-y) - \frac{\partial f}{\partial x_i}(x-y)\right)| \leq |\Gamma(y)| \, \|\frac{\partial f}{\partial x_i}\|_{L^\infty(\mathbb{R}^n)} \mathbb{1}_{B_{R+1}(x)} \in L^1(\mathbb{R}^n)$.

$k \to \infty$, $u_{x_i}(x^k) \to u_{x_i}(x)$

Similarly, $u_{x_ix_j} = \int_{\mathbb{R}^n} \Gamma(y) f_{x_ix_j}(x-y)\,dy$ and continuous.

$\Rightarrow -\Delta u = -\int_{\mathbb{R}^n} \Gamma(y)\,\Delta f(x-y)\,dy = f(x)$. (By property of Γ).

Weak Derivatives

Def1 Suppose $u \in L^1_{loc}(\Omega)$, the distributional derivative $\partial^\alpha u$ can be realised as an $L^1_{loc}(\Omega)$ i.e. $\exists\, v \in L^1_{loc}(\Omega)$ s.t. $\partial^\alpha u = v$ is the distributional sense:

$$\langle \partial^\alpha u, \phi\rangle = (-1)^{|\alpha|}\langle u, \partial^\alpha \phi\rangle = (-1)^\alpha \int_\Omega u\,\partial^\alpha\phi\,dx = (-1)^\alpha \int_\Omega v\phi\,dx. \quad \forall \phi \in C_c^\infty(\Omega)$$

Then we say v is the α-th weak derivative of u write: $v = \partial^\alpha u$

Remark: If $u \in C^{|\alpha|}(\Omega)$, the classical $\partial^\alpha u$ = weak $\partial^\alpha u$

Def2 (k-times weakly differentiable)

$u \in L^1_{loc}(\Omega)$ is k-time weakly differentiable if all $\partial^\alpha u$, $|\alpha| \le k$ exist. $W^k(\Omega) = \{$all such u's$\} \to$ Linear space.

e.g.1 $u(x) = |x|$, $x \in \mathbb{R}$. Guess: $u'(x) = \begin{cases} -1, & x \in (-\infty, 0) \\ 1, & x \in (0, +\infty) \end{cases}$ any value $x = 0$.

$u'(x) \in L^1_{loc}(\Omega)$. $\forall \varphi \in C_c^\infty(\mathbb{R}^n)$. $-\int_{\mathbb{R}} u\varphi'\,dx = -\int_{-\infty}^0 (-x)\varphi'(x)\,dx - \int_0^\infty x\varphi'\,dx$

$= x\varphi\big|_{-\infty}^0 - \int_{-\infty}^0 \varphi(x)\,dx - x\varphi\big|_0^{+\infty} + \int_0^\infty \varphi(x)\,dx = \left(\int_0^\infty - \int_{-\infty}^0\right)\varphi(x)\,dx.$

$\int_{\mathbb{R}} v\varphi\,dx = \int_{-\infty}^0 -\varphi\,dx + \int_0^\infty \varphi\,dx = -\int_{\mathbb{R}} u\varphi'\,dx \Rightarrow v =$ weak u'

Q: Does weak u'' exist? (distri's $u'' = 2\delta_0$) $\to \notin L^1_{loc}(\Omega)$.

If $2\delta_0 \in L^1_{loc}(\mathbb{R})$, $\forall \varphi \in C_c^\infty(\mathbb{R})$ $\langle 2\delta_0, \varphi\rangle = \langle v, \varphi\rangle \Rightarrow 2\varphi(0) = \int_{-\infty}^\infty v\varphi\,dx$

Let $\varphi_\varepsilon(x) = j\left(\frac{x}{\varepsilon}\right)$. standard mollifier.

$\int_{-\infty}^\infty v\varphi_\varepsilon(x)\,dx = \int_{-\infty}^\infty \underline{v\,j\left(\frac{x}{\varepsilon}\right)}\,dx$, Let $\varepsilon \to 0^+$, by DCT, $|v\,j\left(\frac{x}{\varepsilon}\right)| \le |v|\,j(0)\mathbb{1}_{[-1,1]}$

$\to 0 \not\Rightarrow\Leftarrow 2\varphi_\varepsilon(0) = 2j(0) \neq 0.$